COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Analysis	**SECTION:**	CPE21S4
DATE PERFORMED:	July 24, 2026	**DATE SUBMITTED:**	July 24, 2026
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HOA #3 **LINKED LISTS**

1. PROCEDURE OUTPUT

SCREENSHOT

main.cpp | doubly_ll.h

main.cpp > main()

```cpp
#include <iostream>
#include "doubly_ll.h"

int main(){

    // C P E
    DoubleList<char>* head = createNode ('C');
    DoubleList<char>* second = createNode ('P');
    DoubleList<char>* third = createNode ('E');

    // Connect the head node to the next node
    head -> next = second;
    second -> prev = head;
    second -> next = third;
    third -> prev = second;

    // Traverse the list
    std::cout << "Traverse the dllTraverseList: " << std::endl;
    dllTraverse(head);

    // Insert the new node as the head
    std::cout << "Insert a new node as the head." << std::endl;
    dllInsertHead('X', &head);
    dllTraverse(head);

return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Traverse the dllTraverseList:
Forward Traverse: C -> P -> E ->
Reverse Traverse: E -> P -> C ->
Insert a new node as the head.
Forward Traverse: X -> C -> P -> E ->
Reverse Traverse: E -> P -> C -> X ->
Insert a new node at the end.
Forward Traverse: X -> C -> P -> E -> X ->
Reverse Traverse: X -> E -> P -> C -> X ->
```

	
OBSERVATION ANALYSIS	- Just like what we did last time in the Seatwork 3.1, we need to accomplish tasks on making the **TraverseList**, **Insertion**, **General Insertion**, and **Deletion**.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Instructions: These applications have one common property – we traverse the elements of the sequence in a circular fashion. Thus, the node after the last node will be the first node while traversing the list. This is called a circular linked list. We'll take the use case of a music player. It should have following functions supported:

- Create a playlist using multiple songs.
- Add songs to the playlist.
- Remove a song from the playlist.
- Play songs in a loop (for this activity, we will print all the songs once).

SCREENSHOT	

OBSERVATION ANALYSIS	- During my own implementation of the doubly linked list playlist version, I observed that the program was able to perform the required operations (although encountering some errors and requests that I don't usually encounter when working in the lab), including adding songs, removing a selected song, and traversing the playlist in both forward and backward directions. - Since I've almost familiarized it after doing a couple of tries and practices, I now can identify and correct these errors, and the program can be executed as expected.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Asking what's wrong with the execution.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

COMPUTER ENGINEERING

QUEZON CITY



Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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Conclusion:

- As we refresh the lesson we tackled last Wednesday, I came to slowly grasp the concepts and

fundamentals of Linked Listing. So far, I manage to get it (though some minor setbacks in running the code). Overall, making this activity actually made me more engage in doing the activity since it piqued my interest.

Conclusion:

- Although I had already understood the topic from the previous activity, I am still unsure whether the problems I encountered were due to my device or mistakes in my implementation. But nonetheless, this activity further enhance my understanding of linked lists, templates, pointers, and debugging techniques, while demonstrating how proper type consistency and attention to syntax are essential for successful program execution.